(a) You can use raising and lowering operators or first evaluate

$$\langle \ell, m|\, L_x^2 + L_y^2 |\ell, m\rangle \;=\; \langle \ell, m|\, L^2 - L_z^2 |\ell, m\rangle \;=\; \hbar^2\left(\ell(\ell+1) - m^2\right).$$

By symmetry (once one chooses the z-axis the x and y axes can be chosen to be any orthogonal pair in the plane perpendicular to z and so there is nothing to distinguish between them) the two required expectation values are equal and so we have

$$\langle \ell m|L_x^2|\ell m\rangle \;=\; \langle \ell m|L_y^2|\ell m\rangle \;=\; \frac{\hbar^2}{2}\left(\ell(\ell+1) - m^2\right).$$

(b) We have $L_a \;=\; \epsilon_{abc}X_bP_c$ which is just $\vec{L} \;=\; \vec{X}\times\vec{P}$ in component notation. The indices b and c are repeated and are thus assumed to be summed. Use the commutator $[X_b,\, P_c] \;=\; i\hbar\,\delta_{bc}$ and find

$$[L_a,\, X_j] \;=\; \epsilon_{abc}\, X_b\, [P_c,\, X_j] \;=\; -i\hbar\,\epsilon_{abj}X_b \;=\; i\hbar\epsilon_{ajb}X_b\,.$$

Now multiply the constant unit vector n_a and sum over a:

$$[\hat{n}\cdot\vec{L},\, X_a] \;=\; i\hbar\,\epsilon_{ajb}n_aX_b\,.$$

We can use a cyclic permutation of the antisymmetric tensor to write the right-hand side as

$$i\hbar\,\epsilon_{jba}X_bn_a \;=\; i\hbar\,(\vec{X}\times\hat{n})_j\,.$$

We can write this as

$$[\hat{n}\cdot\vec{L},\, \vec{X}] \;=\; i\hbar\,(\vec{X}\times\hat{n})\,.$$

Rotations by α around $\hat{n}$ changes $\vec{X}$ to $\vec{X}'$ given by

$$\vec{X}' \;=\; e^{i\alpha\,\hat{n}\cdot\vec{L}/\hbar}\,\vec{X}\,e^{-i\alpha\,\hat{n}\cdot\vec{L}/\hbar}\,.$$

For infinitesimal rotations we obtain by elementary expansion

$$\vec{X}' \;\approx\; \vec{X} \,+\, \delta\alpha\frac{i}{\hbar}\,[\hat{n}\cdot\vec{L},\, \vec{X}]$$

Using the identity derived earlier $[\hat{n}\cdot\vec{L},\, \vec{X}] \;=\; i\hbar\,(\vec{X}\times\hat{n})\,.$

$$\vec{X}' \;\approx\; \vec{X} \,+\, \delta\alpha\,\hat{n}\times\vec{X}\,.$$

The ordinary position vector transforms exactly this way $\vec{r}' \;=\; \vec{r} \,+\, \delta\alpha\,\hat{n}\times\vec{r}$. This is in Goldstein and Kleppner & Kolenkow.

(c) This makes a mathematical point and is more relevant in electrostatics. Wave functions with discontinuity are not very physical. The function $g(\theta)$ is a discontinuous function: at $\theta = \pi/2$, g jumps from a constant to a zero value. On the other hand all the Legendre polynomials $P_\ell(\cos\theta)$ are continuous functions. A finite sum of continuous functions is continuous. Thus we need an infinite set to represent the given g and therefore, the maximum ℓ is infinite.

(1.2)L_z has eigenvalue $+\hbar$, 0, and $-\hbar$. We order them thus and in the basis of its eigenfunctions L_z is diagonal.[1]

$$L_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

It is easy to verify that the corresponding eigenvectors are

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \text{and} \quad \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

(b) Use the result

$$L_+ \left|1, m\right\rangle = \sqrt{2 - m(m+1)}\hbar \left|1, m+1\right\rangle.$$

The only non-zero matrix elements of L_+ are $\langle 1|L_+|0\rangle$ and $\langle 0|L_+|-1\rangle$, i.e., L_+ raises $|0\rangle$ to $|1\rangle$ and $|-1\rangle$ to $|0\rangle$. The actual values are $\sqrt{2}\,\hbar$ when you substitute $m = 0$ and $m = -1$. Thus we have

$$L_+ = \hbar \begin{pmatrix} 0 & \sqrt{2} & 0 \\ 0 & 0 & \sqrt{2} \\ 0 & 0 & 0 \end{pmatrix}.$$

Clearly L_- is the hermitian conjugate of L_+. We obtain the required matrices using

$$L_x = \frac{1}{2}(L_+ + L_-) \quad \text{and} \quad L_y = \frac{1}{2i}(L_+ - L_-).$$

We have

$$L_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \quad \text{and} \quad L_y = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}.$$

(c) We obtain $\vec{L} \cdot \hat{n}$ by computing

$$L_x \sin\theta\cos\phi + L_y \sin\theta\sin\phi + L_z\cos\theta.$$

[1] Once more with emphasis, given $L_z|m\rangle = m\hbar\,|m\rangle$, it is clear that the mm' matrix element of L_z is given by $\langle m|L_z|m'\rangle = m\hbar\,\langle m|m'\rangle = m\hbar\,\delta_{mm'}$ and is clearly diagonal.

$$\vec{L}\cdot\hat{n} \;=\; \hbar\begin{pmatrix} \cos\theta & \frac{1}{\sqrt{2}}\sin\theta e^{-i\phi} & 0 \\ \frac{1}{\sqrt{2}}\sin\theta e^{i\phi} & 0 & \frac{1}{\sqrt{2}}\sin\theta e^{-i\phi} \\ 0 & \frac{1}{\sqrt{2}}\sin\theta e^{i\phi} & -\cos\theta \end{pmatrix}.$$

You can use *Mathematica* to find its eigenvalues and we find the expected answer $+\hbar$, 0, and $-\hbar$. Verifying that the given column vector is an eigenvector with eigenvalue $+\hbar$ is an exercise in trigonometry. The first element is

$$\begin{aligned} & e^{-i\phi}\cos^2(\theta/2)\cos(\theta) \;+\; e^{-i\phi}\cos(\theta/2)\sin(\theta/2)\sin(\theta) \\ = \;& e^{-i\phi}\,\cos(\theta/2)\,[\,\cos(\theta)\cos(\theta/2)+\sin(\theta)\sin(\theta/2)\,] \\ = \;& e^{-i\phi}\,\cos^2(\theta/2) \end{aligned}$$

reproducing the first element of the given factor. We have used

$$\cos A\,\cos B \,+\, \sin A\,\sin B \;=\; \cos(A-B)\,.$$

The other elements can be checked similarly.

(d) Denote the given state by $|\hat{n}+\rangle$ (it has eigenvalue $+\hbar$ along $\hat{n}$) and the eigenvectors of L_z are $|z+\rangle$, $|z0\rangle$, and $|z-\rangle$ in the notation used in the lectures. The probability of measuring L_z and obtaining $+\hbar$ is $|\,\langle z+|\hat{n}+\rangle\,|^2$ and similarly for the other two eigenvalues of L_z. The results are

$$\cos^4(\theta/2), \quad 2\cos^2(\theta/2)\sin^2(\theta/2), \quad \text{and} \quad \sin^4(\theta/2)$$

for $\hbar$, 0, and $-\hbar$ respectively. **Chapter 5 of Feynman Volume 3 has a discussion of spin one that is instructive.** Compare the answer with the spin-1/2 result shown in class. Compute $\langle S_z\rangle$ and compare with the classical result.

(1.3) Exercise 12.5.3 on page 329. (3 points)

Clearly $\langle \ell, m|J_\pm|\ell, m\rangle \;=\; 0$ since the m value of the ket is either raised from its value of m and becomes orthogonal to the state with m, Result (a) follows.

(b) was done in Problem 1.

(c) The uncertainty principle reads

$$(\Delta J_x)^2 \times (\Delta J_y)^2 \;\geq\; |\langle \ell, m|J_xJ_y|\ell, m\rangle|^2\,.$$

$$(\Delta J_x)^2 \;=\; \langle \ell m|J_x^2|\ell m\rangle^2 \;=\; (\Delta J_y)^2$$

and we have evaluated these in 1(a) and so we obtain

$$\Delta J_x \Delta J_y \;=\; \frac{\hbar^2}{2}\,\Big(j(j+1) \,-\, m^2\Big)\,.$$

We note that

$$J_x J_y \;=\; \frac{J_+ + J_-}{2} \times \frac{J_+ - J_-}{2i} \;=\; \frac{1}{4i}\,\Big(J_+^2 - J_-^2 \,+ J_+ J_- \,-\, J_- J_+\Big)\,.$$

Upon taking expectation values in the state $|\ell, m\rangle$ the first two terms drop out and using the useful commutator $\boxed{[J_+,\, J_-] \;=\; 2\hbar J_z}$ we obtain

$$|\langle \ell, m|J_x J_y|\ell, m\rangle| \;=\; \frac{1}{4}|\langle \ell, m|2\hbar\, J_z|\ell, m\rangle| \;=\; \frac{1}{2}|m|\hbar^2\,.$$

Thus we obtain

$$j(j+1) \,-\, m^2 \,\geq\, |m| \;\Rightarrow\; j(j+1) \,\geq\, |m|(|m|+1).$$

Since $|m| \,\leq\, j$ the inequality is satisfied and when $m \,=\, \pm j$ we have an equality.

(1.4) Part (1) is trivial from 12.5.39 on page 337 and the definition of the Cartesian coordinates in spherical polar coordinates. We have the useful result

$$(x,\, y,\, z) \;=\; \sqrt{\frac{4\pi}{3}}\, r\,\left(-\frac{Y_1^1 - Y_1^{-1}}{\sqrt{2}},\; \frac{i(Y_1^1 + Y_1^{-1})}{\sqrt{2}},\; Y_1^0\right)\,. \qquad (1)$$

(2) We write the wave function as

$$\psi(\vec{r}) \;=\; N\, e^{-\alpha r} r \times \sqrt{\frac{4\pi}{3}}\left(-\frac{Y_1^1 - Y_1^{-1}}{\sqrt{2}} \,+\, \frac{i(Y_1^1 + Y_1^{-1})}{\sqrt{2}} \,+\, 2Y_1^0\right)$$

Thus the coefficients of Y_1^m are $\frac{-1+i}{\sqrt{2}}$, $\frac{1+i}{\sqrt{2}}$,and 2. So the probabilities of finding $m = 1,\, -1$ and 0 are respectively proportional to 1, 1, and 4 (the absolute value squared of the numbers). Normalizing we find the quotes result.

(1.5) Exercise 12.6.9 on page 349. (5 points)

$V(r) \;=\; -V_0$ for $r < r_0$ and vanishes for $r > r_0$. Let

$$k' \;=\; \sqrt{\frac{2\mu(V_0 - W)}{\hbar^2}}\,; \quad \kappa \;=\; \sqrt{\frac{2\mu W}{\hbar^2}}$$

where $W = -E > 0$ and $V_0 - W > 0$.

For $r < r_0$ the radial part of the wave function for $\ell = 0$ obeys the equation (we let $E = -W$)

$$-\frac{\hbar^2}{2\mu}\left(\frac{d^2}{dr^2} + \frac{2}{r}\frac{d}{dr}\right) R_0(r) + \; = \; (-W + V_0)\, R_0(r)\,. \tag{2}$$

$$\frac{d^2 R_0}{dr^2} + \frac{2}{r}\frac{dR_0}{dr} + k'^2 R_0 = 0\,. \tag{3}$$

Using $R_0 = U_0/r$ we have $\frac{d^2 U_0}{dr^2} = -(k')^2 U_0$. The solution for $r < r_0$ is

$$R_0 = A\,\frac{\sin(k'r)}{r}\,.$$

Recall the argument for neglecting the cosine term: For small r, $R_0(r) \propto 1/r$ and $\nabla^2 R_0 \propto \delta(\vec{r})$.

Similarly outside we have the equation $U_0'' = \kappa^2 U_0$ and the solution for $r > r_0$ is

$$R_0 = B\,\frac{e^{-\kappa r}}{r}\,.$$

Equating logarithmic derivatives (dropping constants) we have

$$\frac{d}{dr}\left(-\log r + \log(\sin(k'r))\right)\Big|_{r=r_0} = \frac{d}{dr}\left(-\log r - \kappa r\right)\Big|_{r=r_0}$$

So we obtain simply

$$-\kappa = k' \cot(k' r_0)$$

the result in the book. Since $\kappa > 0$ for exponential decay the cotangent has to be negative which means that its argument must be greater than $\pi/2$. Thus

$$k' > \frac{\pi}{2r_0}\,.$$

Now the energy can be evaluated inside the well to be

$$E = \frac{\hbar^2 k'^2}{2\mu} - V_0 \le 0$$

since we have a bound state. Thus

$$V_0 > \frac{\hbar^2 k'^2}{2\mu} \text{ and since } k' > \frac{\pi}{2r_0} \text{ we have } V_0 > \frac{\hbar^2\pi^2}{8\mu r_0^2}\,.$$

Otherwise, there are no bound states.

(1.6) Consider a quantum mechanical top. Explain clearly what is meant by a top classically and write down the quantum Hamiltonian for an asymmetrical top and justify it carefully. The quantum mechanics of tops is tricky conceptually.

For a careful discussion if you ever need it please consult Landau and Lifshitz "Quantum Mechanics".

We take the correspondence principle point of view, write down the classical hamiltonian and then quantize it according to the postulates. Note that the moments of inertia along the space-fixed axes are time-dependent. so we use body-fixed axes. Let the body-fixed coordinates be denoted by u, v, and w and the corresponding unit vectors $\hat{u}$, $\hat{v}$, and $\hat{w}$. Note that these are time-dependent in the space-fixed axes. The corresponding angular momenta are denoted by L_u, L_v, and L_w and the classical Hamiltonian can be "quantized" to obtain

$$H = \frac{L_u^2}{2I_u} + \frac{L_v^2}{2I_v} + \frac{L_w^2}{2I_w}$$

where the principal moments of inertia are time-independent.

When we quantize the commutation relations of the angular momentum operators in the rotating frame are not obvious since they are projected along time-dependent axes. A careful calculation yields

$$[L_u, L_v] = -i\hbar L_w$$

with the "wrong" sign. The entire derivation of the eigenvalues goes through the same way even with the wrong sign!

In the next two parts we will consider two Hamiltonians with a similar algebraic structure.

For the Hamiltonian $H_0 = \vec{L}\cdot\vec{L}/(2I)$ the energy eigenvalues are $\frac{\hbar^2}{2I}\ell(\ell+1)$ with degeneracy $2\ell + 1$ and with spherical harmonic eigenfunctions.

For $H_1 = \frac{1}{2I_0}(L_x^2 + L_y^2) + \frac{1}{2I_z}L_z^2$ using $L^2 - L_z^2 = L_x^2 + L_y^2$ the eigenvalues are

$$\hbar^2 \left[\frac{1}{2I_0}\ell(\ell+1) + \left(\frac{1}{2I_z} - \frac{1}{2I_0}\right) m^2\right].$$

For a given ℓ, the two states with quantum numbers $\pm m$ yield the same energy and thus a double degeneracy except for $m = 0$. *The degeneracy of the symmetrical top is more complicated.*